
Complements of Operational Research

Exercise sheet 3 — Sequencing problems (Johnson, TDM, *branch and bound*)

The exercises marked **homework** are to be solved at home and sent by email to the lecturer, by the deadline given in class; the solution is shared afterwards. The others are from previous tests and exams. In all the problems with several machines, the jobs go through the machines in the same order and each machine processes one job at a time.

Exercise 1 — Total tardiness, three jobs

Homework (Exercise 1)

Determine, by the *branch and bound* method, the sequence of jobs that minimises the total tardiness in the completion of all the jobs.

Job	Processing time (days)	Due date (days from today)
A	4	6
B	5	13
C	7	8

Exercise 2 — Total tardiness, four jobs

Homework (Exercise 2)

Same statement as the previous exercise, with the following table.

Job	Processing time (days)	Due date (days from today)
A	3	4
B	6	10
C	5	7
D	4	6

Exercise 3 — Maintenance jobs in a workshop

Homework (1st test, 27/03/2025)

A workshop has four maintenance jobs (P, Q, R, S) to carry out. Each job has a processing time and a due date (in days, counted from today). We want to determine the optimal sequence in which to carry out the jobs, minimising the accumulated total tardiness. Solve the problem by the *branch and bound* method.

Job	Processing time (days)	Due date (days from today)
P	4	5
Q	3	8
R	5	9
S	2	6

- What is the optimal sequence of the jobs that minimises the accumulated total tardiness?
- Give the value of the minimum total tardiness associated with the optimal sequence obtained.
- What is the total time needed to complete all the jobs in that sequence?

Exercise 4 — Engine workshop: when Johnson's condition fails

Homework (new)

A workshop repairs four engines (A, B, C, D). Each one goes, in this order, through disassembly (machine 1), machining (machine 2) and assembly (machine 3). The times, in hours, are:

Engine	A	B	C	D
Disassembly (M1)	8	3	3	2
Machining (M2)	4	9	9	4
Assembly (M3)	5	2	8	9

- Show that the condition that allows Johnson's method to be applied to three machines does not hold.
- Even so, apply Johnson's method to the two fictitious machines ($G = M_1 + M_2$, $H = M_2 + M_3$) and the TDM. Compute the total time of each sequence.
- Determine the optimal sequence by the *branch and bound* method (lower bound for each machine, as in class). How much is gained with respect to the sequences of part b)?

Exercise 5 — Johnson, four machines

1st test, 13/04/2018

A company has to plan the optimal sequence of 5 jobs. Each of the jobs is known to require the use of four machines sequentially. The expected times for carrying out each of the jobs on the four machines are given in the following table:

Job	t1	t2	t3	t4	t5
Machine 1	11	13	9	15	17
Machine 2	5	2	5	5	4
Machine 3	7	6	5	3	8
Machine 4	15	8	13	10	11

If possible, solve the problem with Johnson's method. Give the total time to carry out all the jobs and the "slack" (idle) times of the four machines.

Exercise 6 — Johnson, three machines

exam, 10/07/2018

A company has to plan the optimal sequence of 5 jobs. Each of the jobs is known to require the use of three machines sequentially. The expected times for carrying out each of the jobs on the three machines are given in the following table:

Job	t1	t2	t3	t4	t5
Machine 1	9	10	6	8	10
Machine 2	5	6	2	3	4
Machine 3	6	9	7	5	6

If possible, solve the problem with Johnson's method. Give the total time to carry out all the jobs and the "slack" (idle) times of the three machines.

Exercise 7 — Johnson, four machines

1st test, 08/04/2019

A metalworking company has to plan the optimal sequence of 5 jobs. Each of the jobs is known to require the use of four machines sequentially. The expected times for carrying out each of the jobs on the four machines are given in the following table:

Job	t1	t2	t3	t4	t5
Machine 1	12	13	11	15	16
Machine 2	4	3	5	2	6
Machine 3	6	7	5	8	4
Machine 4	15	10	9	8	11

If possible, solve the problem with Johnson's method. Give the total time to carry out all the jobs and the "slack" (idle) times of the four machines.

Exercise 8 — Johnson, three machines

exam, 27/06/2019

A company has to plan the optimal sequence of 5 jobs. Each of the jobs is known to require the use of three machines sequentially. The expected times for carrying out each of the jobs on the three machines are given in the following table:

Job	t1	t2	t3	t4	t5
Machine 1	8	12	15	9	10
Machine 2	8	5	7	6	4
Machine 3	9	7	5	12	10

If possible, solve the problem with Johnson's method. Give the total time to carry out all the jobs and the "slack" (idle) times of the three machines.

Exercise 9 — Johnson, four machines

exam, 11/07/2019

A company has to plan the optimal sequence of 5 jobs. Each of the jobs is known to require the use of four machines sequentially. The expected times for carrying out each of the jobs on the four machines are given in the following table:

Job	t1	t2	t3	t4	t5
Machine 1	12	13	9	16	17
Machine 2	6	2	5	3	4
Machine 3	4	7	5	8	6
Machine 4	14	8	13	9	11

If possible, solve the problem with Johnson's method. Give the total time to carry out all the jobs and the "slack" (idle) times of the four machines.

Exercise 10 — Johnson, three machines

exam, 26/07/2019

A company has to plan the optimal sequence of 5 jobs. Each of the jobs is known to require the use of three machines sequentially. The expected times for carrying out each of the jobs on the three machines are given in the following table:

Job	t1	t2	t3	t4	t5
Machine 1	9	10	6	8	11
Machine 2	3	5	6	5	4
Machine 3	5	9	8	6	4

If possible, solve the problem with Johnson's method. Give the total time to carry out all the jobs and the "slack" (idle) times of the three machines.

Exercise 11 — Johnson, four machines

exam, 05/07/2021

A metalworking company has to plan the optimal sequence of 5 jobs. Each of the jobs is known to require the use of four machines sequentially. The expected times for carrying out each of the jobs on the four machines are given in the following table:

Job	t1	t2	t3	t4	t5
Machine 1	12	13	11	15	16
Machine 2	4	4	5	2	6
Machine 3	6	7	5	8	4
Machine 4	15	10	9	8	11

If possible, solve the problem with Johnson's method. Give the total time to carry out all the jobs and the "slack" (idle) times of the four machines.

Exercise 12 — TDM, four machines

1st test, 03/04/2024

A company has to plan the optimal sequence of 5 jobs. Each of the jobs is known to require the use of four machines sequentially. The expected times for carrying out each of the jobs on the four machines are given in the following table:

Job	t1	t2	t3	t4	t5
Machine 1	17	13	9	11	11
Machine 2	5	2	4	5	4
Machine 3	7	6	5	3	8
Machine 4	11	7	13	10	15

If possible, solve the problem with the *Time Deviation Method* (TDM). Give the total time to carry out all the jobs and the "slack" (idle) times of the four machines.

Exercise 13 — TDM, four machines

exam, 02/07/2024

A company has to plan the optimal sequence of 5 jobs. Each of the jobs is known to require the use of four machines sequentially. The expected times for carrying out each of the jobs on the four machines are given in the following table:

Job	t1	t2	t3	t4	t5
Machine 1	10	17	9	11	9
Machine 2	7	2	4	5	4
Machine 3	5	2	5	6	8
Machine 4	10	11	13	13	10

If possible, solve the problem with the *Time Deviation Method* (TDM). Give the total time to carry out all the jobs and the "slack" (idle) times of the four machines.